

Total No. of Questions : 8]

SEAT No. :

P-627

[Total No. of Pages : 2

[6004]-581

B.E. (Information Technology)

AUGMENTED AND VIRTUAL REALITY

(2019 Pattern) (Semester - VIII) (414452B) (Elective - VI)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) *Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6 and Q.7 or Q.8.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Assume Suitable data if necessary.*

Q1) a) Explain Perception of colors in detail. [9]

b) Describe different motion detection mechanism in detail. [9]

OR

Q2) a) Explain Ray Tracing and Shading Models in detail. [9]

b) Explain Three categories of tracking in VR systems. [9]

Q3) a) How does Augmented Reality Work? Explain with suitable example. [9]

b) Describe different applications of Augmented Reality. [8]

OR

Q4) a) Define Augmented Reality? Explain History of Augmented Reality.[8]

b) Describe different hardware components used in augmented reality.[9]

Q5) a) What is hybrid tracking? Explain with suitable examples. [9]

b) What is role of sensor in tracking? Explain in detail. [9]

OR

Q6) a) List and explain characteristics of Tracking Technology. [9]

b) Describe Spatial Display Model in detail. [9]

P.T.O.

- Q7)** a) What is natural-feature tracking? Describe with any one application.[9]
b) What is simultaneous localization and mapping in AR? [8]

OR

- Q8)** a) What software is used for augmented reality? Explain in detail. [9]
b) What is marker tracking in AR? Explain in detail. [8]

